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### Vibration feedback device, handheld communication device, and wearable device

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#### Abstract

A vibration feedback device, a handheld communication device including the same and a wearable device. The vibration feedback device is provided on a frame and includes a button fixed to the frame; a sensor for detecting a pressing state of the button; and a vibration motor fixed to the frame and arranged below the button. The frame includes an outer side surface, an inner side surface, and a mounting hole penetrating through the outer and inner side surfaces; the vibration motor includes a stator fixed to the inner side surface, an antenna arranged in the mounting hole, and an elastic connection member connecting the antenna to the stator; and a direction of a pressing force of the button is perpendicular to a vibrating direction of the antenna. The vibration feedback device provides a vibration feedback that is different from the direction of the pressing force and is diversified.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/CN2023/096708, filed on May 29, 2023, which hereby is incorporated by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates to the technical field of vibration feedback, and in particular, to a vibration feedback device, a handheld communication device including the vibration feedback device, and a wearable device.

BACKGROUND

(2) The vibration feedback technology can reproduce a touch feeling for the user through a series of actions such as acting force and vibration. This mechanical stimulus may be applied to assisted creation and control of virtual scenes or virtual objects in computer simulations, as well as to enhance remote manipulation of machines and apparatus.

(3) However, the existing vibration feedback product generally has an insufficient tactile feedback precision and a single vibration feeling, which affect the user experience.

(4) In view of this, the present disclosure provides a new vibration feedback device to solve the above-mentioned technical problems.

SUMMARY

(5) An object of the present disclosure is to provide a vibration feedback device, a vibrating direction of which is different from a direction of a pressing force and is diversified.

(6) In an aspect, the present provides a vibration feedback device, the vibration feedback device is

provided on a frame, and includes: a button fixed to the frame; a sensor for detecting a pressing state of the button; and a vibration motor fixed to the frame and arranged below the button. The frame has a mounting hole penetrating through the frame; the frame includes an outer side surface and an inner side surface opposite to the outer side surface, and the mounting hole penetrates through both the outer side surface and the inner side surface; the vibration motor includes a stator fixed to the inner side surface of the frame, an antenna arranged in the mounting hole, and an elastic connection member that connects the antenna to the stator; and a direction of a pressing force of the button is perpendicular to a vibrating direction of the antenna.

(7) As an improvement, the stator includes a baseplate fixed to the frame and a solenoid assembly fixed to the baseplate, and the elastic connection member includes an end connected to the antenna and another end connected to the baseplate.

(8) As an improvement, the baseplate includes a bearing portion for bearing the solenoid assembly, a first extension portion extending from the bearing portion to two sides and connected to the frame, and a second extension portion extending from the bearing portion to the mounting hole and connected to the elastic connection member.

(9) As an improvement, the antenna includes a housing and a magnet fixed in the housing, the housing is fixed to the button, and the magnet is arranged at two sides of the solenoid assembly.

(10) As an improvement, the housing includes a top wall and a side wall bent and extending from the top wall, and the top wall and the side wall jointly define a receiving space; and the top wall is fixed relative to the button, and the magnet is fixed to the side wall.

(11) As an improvement, the housing is made of a magnetic conductive material.

(12) As an improvement, the sensor is of a sheet structure and is clamped between the top wall and the button.

(13) As an improvement, the magnet includes a pair of magnets, the pair of magnets is magnetized along the vibrating direction of the antenna, and the pair of magnets is arranged with same poles facing each other.

(14) In another aspect, the present disclosure provides a handheld communication device, including a frame, a screen fixed to a side of the frame, and a backplate fixed to another side of the frame. The handheld communication device further includes a vibration feedback device fixed to the frame, and the vibration feedback device includes: a button fixed to the frame; a sensor for detecting a pressing state of the button; and a vibration motor fixed to the frame and arranged below the button. The frame has a mounting hole penetrating through the frame; the frame includes an outer side surface and an inner side surface opposite to the outer side surface, and the mounting hole penetrates through both the outer side surface and the inner side surface; the vibration motor includes a stator fixed to the inner side surface of the frame, an antenna arranged in the mounting hole, and an elastic connection member that connects the antenna to the stator; and a direction of a pressing force of the button is perpendicular to a vibrating direction of the antenna.

(15) In another aspect, the present disclosure provides a wearable device, including a rim, a lens provided on the rim, and a frame extending backwards from two sides of the rim. The wearable device further includes a vibration feedback device fixed to the frame, and the vibration feedback device includes: a button fixed to the frame; a sensor for detecting a pressing state of the button; and a vibration motor fixed to the frame and arranged below the button. The frame has a mounting hole penetrating through the frame; the frame includes an outer side surface and an inner side surface opposite to the outer side surface, and the mounting hole penetrates through both the outer side surface and the inner side surface; the vibration motor includes a stator fixed to the inner side surface of the frame, an antenna arranged in the mounting hole, and an elastic connection member that connects the antenna to the stator; and a direction of a pressing force of the button is perpendicular to a vibrating direction of the antenna.

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# Description

## BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a perspective view of a vibration feedback device according to an embodiment of the present disclosure;
- (2) FIG. 2 is an exploded view of the vibration feedback device shown in FIG. 1;
- (3) FIG. 3 is a cross-sectional view taken along A-A when the vibration feedback device shown in FIG. 1 is provided on the frame;
- (4) FIG. 4 is a perspective view of a handheld communication device according to an embodiment of the present disclosure;
- (5) FIG. 5 is a partially exploded view of the handheld communication device shown in FIG. 4; and
- (6) FIG. 6 is a partially exploded view of a wearable device according to an embodiment of the present disclosure.

## DESCRIPTION OF EMBODIMENTS

(7) The technical solutions in embodiments of the present disclosure will be described below in connection with the drawings in the embodiments of the present disclosure, and it is apparent that the embodiments described herein are merely a part, not all of the embodiments of the present disclosure. Based on the embodiments of the present disclosure, all other embodiments obtained by those of ordinary skill in the art without creative efforts shall fall within a scope of the present disclosure.

(8) Referring to FIG. 1 to FIG. 3, an embodiment of the present disclosure provides a vibration feedback device **100** provided on a frame **101**, and the vibration feedback device **100** includes a button **1** fixed to the frame **101**, a sensor **2** for detecting a pressing state of the button, and a vibration motor **3** fixed to the frame **101** and arranged below the button **1**.

(9) The frame **101** has a mounting hole **1011** penetrating through the frame **101**. The frame **101** includes an outer side surface **1012** and an inner side surface **1013** opposite to the outer side surface **1012**, and the mounting hole **1011** penetrates through both the outer side surface **1012** and the inner side surface **1013**. The button **1** protrudes from the outer side surface **1012** of the frame **101**. The outer side surface **1012** has a recess **1014**, and the inner side surface **1013** has a protrusion **1015**. The button **1** is fitted and fixed in the recess **1014** of the outer side surface **1012**, and the vibration motor **3** is fixedly connected to the protrusion **1015**.

(10) In this embodiment, the sensor **2** is of a sheet structure and is arranged between the button **1** and the vibration motor **3**. In other embodiments of the present disclosure, the sensor **2** may have another shape/be of another structure and may be mounted at another position of the vibration feedback device **100**. The sensor **2** is configured to detect position information and strength information of a pressing force received by the button **1**.

(11) The vibration motor **3** includes a stator **31** fixed to the inner side surface **1013** of the frame **101**, an antenna **32** located in the mounting hole **1011**, and an elastic connection member **33** that connects the antenna **32** to the stator **31**. The antenna **32** vibrates in a left-right direction, and a direction of the pressing force of the button **1** is perpendicular to a vibrating direction of the antenna **32**.

(12) The stator **31** includes a baseplate **311** fixed to the frame **101** and a solenoid assembly **312** fixed to the baseplate **311**, and the baseplate **311** is fixedly connected to the protrusion **1015** of the frame **101**. The elastic connection member **33** includes an end connected to the antenna **32**, and another end connected to the baseplate **311**. For example, the baseplate **311** includes a bearing portion **3111** for bearing the solenoid assembly **312**, a first extension portion **3112** extending from the bearing portion **3111** towards two sides and fixedly connected to the frame **101**, and a second extension portion **3113** extending from the bearing portion **3111** towards the mounting hole **1011** and connected to the elastic connection member **33**. The solenoid assembly **312** includes an iron

core **3121** made of a magnetic conductive material, and a solenoid **3122** wound around the iron core **3121**.

(13) The antenna **32** includes a housing **321** fixed relative to the button **1**, and a magnet **322** fixed in the housing **321**, and the magnet **322** is arranged at two sides of the solenoid assembly **312**. The housing **321** includes a top wall **3211** fixed relative to the button **1**, and a side wall **3212** bent and extending from the top wall **3211**. The top wall **3211** the side wall **3212** jointly define a receiving space, and the magnet **322** is fixed to the side wall **3212**. The housing **321** is made of a magnetic conductive material. The sensor **2** is clamped between the top wall **3211** and the button **1**. The magnet **322** includes a pair of magnets **322**. Along the vibrating direction of the antenna, the pair of magnets **322** is magnetized, and the pair of magnets **322** is arranged with same poles facing each other.

(14) The elastic connection member **33** is of a flat sheet structure, and the elastic connection member **33** is made of a metal material.

(15) During normal operation, when a finger presses the button **1**, a pressure is generated in a direction perpendicular to the surface of the button **1**, then the elastic connection member **33** deforms, and the sensor **2** detects action information of the pressure through a deformation of the elastic connection member **33**. The action information includes a position information and a strength information of the pressure. A corresponding electrical signal is provided to the solenoid **3122** of the vibration motor **3** through a processor and a signal algorithm, so that the antenna **32** of the vibration motor **3** vibrates in a direction perpendicular to the pressure of the button, thereby providing a tactile feedback to the user. The tactile effect may be a button effect, a sliding button effect, and the like.

(16) As shown in FIG. 4 to FIG. 5, an embodiment of the present disclosure further provides a handheld communication device **200**, including a frame **101**, a screen **201** fixed to a side of the frame **101**, a backplate **202** fixed to another side of the frame **101**, and the vibration feedback device **100** described above. The vibration feedback device **100** is provided in a mounting hole **1011** of the frame **101** to provide a tactile feedback.

(17) As shown in FIG. 6, an embodiment of the present disclosure further provides a wearable device **300** including a rim **301**, a lens **302** provided on the rim **301**, a frame **101** extending backwards from two sides of the rim **301**, and the vibration feedback device **100** described. The vibration feedback device **100** is provided in a mounting hole **1011** of the frame **101** to provide a tactile feedback. The frame **101** includes two components, that is, two legs of the wearable device **300**. Each of the two legs is provided with the vibration feedback device **100**. The wearable device **300** further includes a cover **303** that covers the mounting hole **1011** of the frame **101** to encapsulate the vibration feedback device **100**.

(18) Compared with the related art, the vibration feedback device provided by the present disclosure is provided on a frame and includes a button fixed to the frame, a sensor for detecting a pressing state of the button, and a vibration motor fixed to the frame and arranged below the button; the frame has a mounting hole penetrating through the frame and includes an outer side surface and an inner side surface opposite to the outer side surface, the mounting hole penetrates through both the outer side surface and the inner side surface; the vibration motor includes a stator fixed to the inner side surface of the frame, an antenna arranged in the mounting hole, and an elastic connection member that connects the antenna to the stator; and a direction of the pressing force of the button is perpendicular to a vibrating direction of the antenna. The present disclosure further provides a handheld communication device and a wearable device including the vibration feedback device. The vibrating direction of the vibration feedback device is perpendicular to the direction of the pressing force, thereby providing a different tactile vibration for users, and the vibration feeling is diversified. The handheld communication device and the wearable device including the vibration feedback device provided by the present disclosure further have a diverse tactile feedback.

(19) The above description merely describes some of, rather than all of embodiments of the present disclosure, and it can be understood that those skilled in the art can further make improvements without departing from a concept of the present disclosure, but all of these improvements shall fall within a scope of the present disclosure.

## Claims

1. A vibration feedback device, wherein the vibration feedback device is provided on a frame, and comprises: a button fixed to the frame; a sensor for detecting a pressing state of the button; and a vibration motor fixed to the frame and arranged below the button, wherein the frame has a mounting hole penetrating through the frame; the frame comprises an outer side surface and an inner side surface opposite to the outer side surface, and the mounting hole penetrates through both the outer side surface and the inner side surface; the vibration motor comprises a stator fixed to the inner side surface of the frame, an antenna arranged in the mounting hole, and an elastic connection member that connects the antenna to the stator; and a direction of a pressing force of the button is perpendicular to a vibrating direction of the antenna; the stator comprises a baseplate fixed to the frame and a solenoid assembly fixed to the baseplate, and the elastic connection member comprises an end connected to the antenna and another end connected to the baseplate; the antenna comprises a housing and a magnet fixed in the housing, the housing is fixed to the button, and the magnet is arranged at two sides of the solenoid assembly; the magnet comprises a pair of magnets, the pair of magnets is magnetized along the vibrating direction of the antenna, and the pair of magnets is arranged with same poles facing each other.
2. The vibration feedback device as described in claim 1, wherein the baseplate comprises a bearing portion for bearing the solenoid assembly, a first extension portion extending from the bearing portion to two sides and connected to the frame, and a second extension portion extending from the bearing portion to the mounting hole and connected to the elastic connection member.
3. The vibration feedback device as described in claim 1, wherein the housing comprises a top wall and a side wall bent and extending from the top wall, and the top wall and the side wall jointly define a receiving space; and the top wall is fixed relative to the button, and the magnet is fixed to the side wall.
4. The vibration feedback device as described in claim 1, wherein the housing is made of a magnetic conductive material.
5. The vibration feedback device as described in claim 3, wherein the sensor is of a sheet structure and is clamped between the top wall and the button.
6. A handheld communication device, comprising a frame, a screen fixed to a side of the frame, and a backplate fixed to another side of the frame, wherein the handheld communication device further comprises a vibration feedback device fixed to the frame, and the vibration feedback device comprises: a button fixed to the frame; a sensor for detecting a pressing state of the button; and a vibration motor fixed to the frame and arranged below the button, wherein the frame has a mounting hole penetrating through the frame; the frame comprises an outer side surface and an inner side surface opposite to the outer side surface, and the mounting hole penetrates through both the outer side surface and the inner side surface; the vibration motor comprises a stator fixed to the inner side surface of the frame, an antenna arranged in the mounting hole, and an elastic connection member that connects the antenna to the stator; and a direction of a pressing force of the button is perpendicular to a vibrating direction of the antenna; the stator comprises a baseplate fixed to the frame and a solenoid assembly fixed to the baseplate, and the elastic connection member comprises an end connected to the antenna and another end connected to the baseplate; the antenna comprises a housing and a magnet fixed in the housing, the housing is fixed to the button, and the magnet is arranged at two sides of the solenoid assembly; the magnet comprises a pair of magnets, the pair of magnets is magnetized along the vibrating direction of the antenna, and the pair of magnets is

arranged with same poles facing each other.

7. A wearable device, comprising a rim, a lens provided on the rim, and a frame extending backwards from two sides of the rim, wherein the wearable device further comprises a vibration feedback device fixed to the frame, and the vibration feedback device comprises: a button fixed to the frame; a sensor for detecting a pressing state of the button; and a vibration motor fixed to the frame and arranged below the button, wherein the frame has a mounting hole penetrating through the frame; the frame comprises an outer side surface and an inner side surface opposite to the outer side surface, and the mounting hole penetrates through both the outer side surface and the inner side surface; the vibration motor comprises a stator fixed to the inner side surface of the frame, an antenna arranged in the mounting hole, and an elastic connection member that connects the antenna to the stator; and a direction of a pressing force of the button is perpendicular to a vibrating direction of the antenna; the stator comprises a baseplate fixed to the frame and a solenoid assembly fixed to the baseplate, and the elastic connection member comprises an end connected to the antenna and another end connected to the baseplate; the antenna comprises a housing and a magnet fixed in the housing, the housing is fixed to the button, and the magnet is arranged at two sides of the solenoid assembly; the magnet comprises a pair of magnets, the pair of magnets is magnetized along the vibrating direction of the antenna, and the pair of magnets is arranged with same poles facing each other.

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